Flywheels

from zero to wheel

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Examples

Figure: Flywheel of extracting machine in Luiza Mine, Zabrze (source: `w.wiki/CpZ3`)

Figure: Potter's wheel (source: `w.wiki/CpZ4`)

Figure: Fidget spinner (source: `w.wiki/CpZ5`)

Energy storage technologies

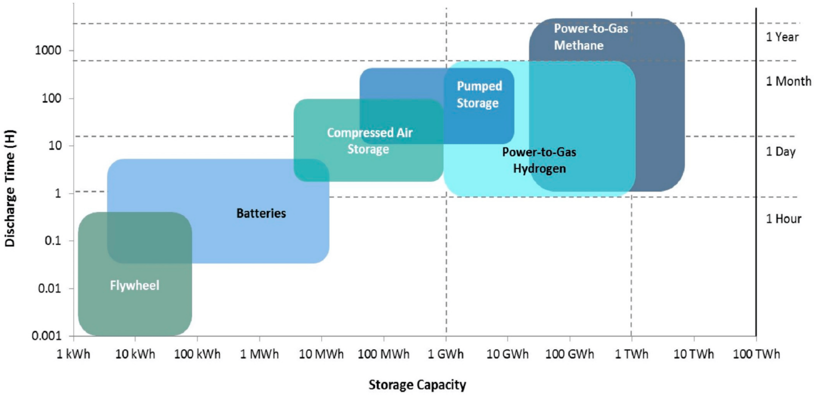


Figure: Comparison of discharge time vs capacity of energy storage technologies (source: `doi.org/10.3390/en9090674`)

Why flywheels are cool

Advantages:

- High Efficiency
- Durability
- Fast Response Time
- Environmentally Friendly
- High Power Density

Disadvantages:

- Self-Discharge
- Material Constraints
- High Initial Cost
- Low Energy Density
- Complexity of Bearings

Physical characteristics

moment of inertia: $J = \int\limits_m r^2 \, \mathrm{d}m$

rotational energy: $E_{kin} = \frac{1}{2} J \omega^2$

Specific energy

$$\frac{E}{m} = K\frac{\sigma}{\rho}$$

- K – shape factor [dimensionless]
- σ – tensile strength [Pa]

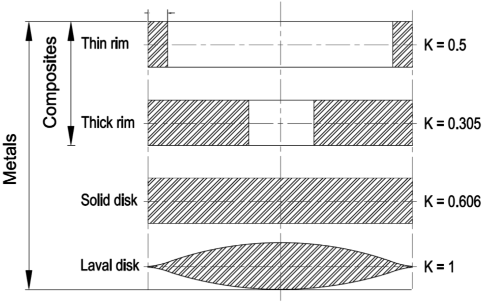


Figure: Different flywheel cross sections and their shape factors (source: doi.org/10.3390/app7030286)

Grid storage

Figure: Beacon Power's frequency regulation plant in Hazle, PA (source: doi.org/10.1109/IECON.2018.8591842)

Transportation

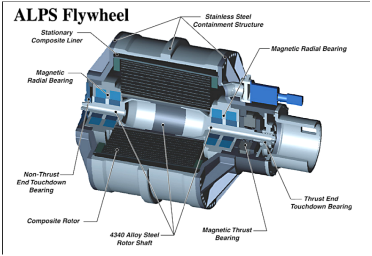


Figure: Section view of the ALPS (Advanced Locomotive Propulsion System) composite flywheel (source: `doi.org/10.1109/VETECF.2003.1286244`)

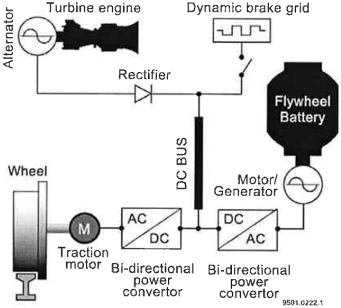


Figure: ALPS schematic diagram (source: `hdl.handle.net/2152/33077`)

Space applications

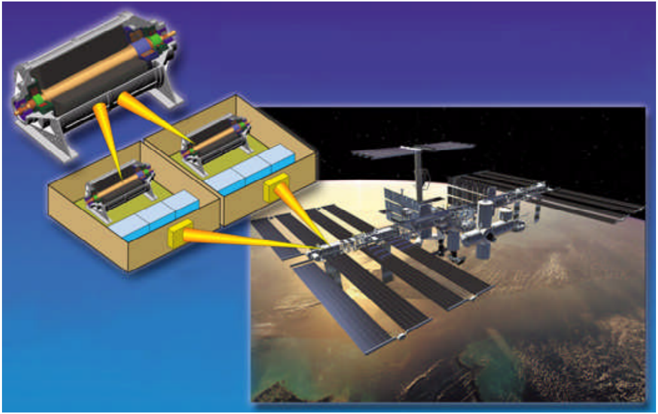
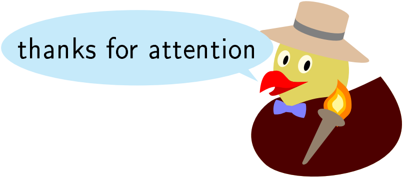


Figure: Flywheel Energy Storage System (FESS) architecture for the International Space Station (source: `doi.org/10.1109/6.993788`)

Comparison

Table: Flywheel applications comparison (source: hdl.handle.net/2152/33077)

Parameter	ALPS System	Transit Bus System	NASA Space Station
Energy Stored (kW-hr)	133	2	3.6
Peak Power (kW)	2,200	150	3.6
Typical Discharge Time	3 minutes	30 seconds	30 minutes
Rotational Speed (rpm)	15,000	40,000	53,000
Machine Weight (lbs)	19,000	450	250
Motor/Generator	Induction	Permanent Magnet	Permanent Magnet
Cooling	Air and Oil	Oil and Water	Cold Plate
Bearings	Homopolar Magnetic	Homopolar Magnetic	Homopolar Magnetic
Flywheel Design	UT-CEM Cylindrical	UT-CEM Cylindrical	UT-CEM Cylindrical
Rotor Tip Speed (m/s)	950	935	920



https://github.com/falcotinnunculus/przezrocza/blob/wladca/red_energy/flywheel.pdf